The University of Birmingham
School of Computer Science

Second Exercise Sheet for Week 5 - Abstract data types

Question 1: Type-checking with existential types

Give a derivation to show that each of the following is a valid typing judgement.

(a) $\vdash \texttt{pack}\ \langle \mathbb{N}\times\mathbb{N}, \lambda p : \mathbb{N}\times\mathbb{N}.\langle \texttt{snd}\, p, \texttt{fst}\, p\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha \to \alpha : \exists\alpha.\alpha\to\alpha$

(b) $\vdash \texttt{pack}\ \langle \mathbb{N}, \lambda p : \mathbb{N}\times\mathbb{N}.\langle \texttt{snd}\, p, \texttt{fst}\, p\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha\times\alpha \to \alpha\times\alpha : \exists\alpha.\alpha\times\alpha\to\alpha\times\alpha$

(c) $\vdash \texttt{unpack}\ \alpha, x = P\ \texttt{in}\ (\texttt{snd}\, x)\,(\texttt{fst}\, x)\,\texttt{zero} : \mathbb{B}$
where $P = \texttt{pack}\ \langle \mathbb{N}, \langle \texttt{zero}, \lambda x : \mathbb{N}.\lambda y : \mathbb{N}.\,\texttt{true}\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha\times(\alpha\to\mathbb{N}\to\mathbb{B})$.

(d) $m : \exists\alpha.\alpha\times(\alpha\to\alpha)\times(\alpha\to\mathbb{B}) \vdash \texttt{unpack}\ \gamma, p = m\ \texttt{in}\ (\texttt{snd}\, p)\,\big((\texttt{snd}\,(\texttt{fst}\, p))\,(\texttt{fst}\,(\texttt{fst}\, p))\big) : \mathbb{B}$

Question 2: 'Type inference' with existential types

For each of the following incomplete typing judgements in (Church-style) System F extended with (pairs and) existential types, fill in the holes where a missing type expression should be. (There may be multiple possible answers).

(a) $\vdash \texttt{pack}\ \langle \mathbb{B}\times\mathbb{N}, \langle \lambda x : \mathbb{B}.x, \lambda y : \mathbb{N}.\langle \texttt{iszero}\, y, y\rangle\rangle\rangle\ \texttt{as}\ \underline{??} : \underline{??}$

(b) $\vdash \texttt{pack}\ \langle \forall\alpha.\alpha\to\alpha, \lambda f : \forall\alpha\to\alpha.\alpha.\ \lambda x : \mathbb{N}.f\{\mathbb{N}\}\,x\rangle\ \texttt{as}\ \underline{??} : \underline{??}$

(c) $f : \forall\alpha.\alpha\to\alpha\to\alpha \vdash \texttt{unpack}\ \theta, x = \texttt{pack}\ \langle \mathbb{N}, \langle \lambda n : \mathbb{N}.\texttt{pred}\, n, \lambda n : \mathbb{N}.\texttt{iszero}\, n\rangle\rangle\ \texttt{as}\ \underline{??}\ \texttt{in}$
$\texttt{snd}\, x\,(f\{\theta\}\,(\texttt{fst}\, x\,\texttt{zero})\,(\texttt{fst}\, x\,\texttt{one})) : \underline{??}$

(d) $O : \underline{??} \vdash \texttt{unpack}\ \alpha, x = O\ \texttt{in}\ \texttt{pack}\ \langle \alpha, \langle \texttt{if}\ \texttt{fst}\, x\ \texttt{then}\ \texttt{false}\ \texttt{else}\ \texttt{true}, \texttt{snd}\, x\rangle\rangle\ \texttt{as}\ \underline{??} : \underline{??}$

Question 3: Abstraction barriers

Determine whether each of the following terms is typeable.

(a) $\texttt{unpack}\ \nu, x = \texttt{pack}\ \langle \mathbb{N}, \langle \texttt{one}, \lambda x : \mathbb{N}.\,\texttt{succ}\, x\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha\times(\alpha\to\alpha)\ \texttt{in}\ \texttt{succ}\,(\texttt{fst}\, x)$

(b) $\texttt{unpack}\ \nu, x = \texttt{pack}\ \langle \mathbb{N}, \langle \texttt{one}, \lambda x : \mathbb{N}.\,\texttt{succ}\, x\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha\times(\alpha\to\alpha)\ \texttt{in}\ (\texttt{snd}\, x)\,(\texttt{fst}\, x)$

(c) $\texttt{unpack}\ \nu, x = \texttt{pack}\ \langle \mathbb{N}, \langle \texttt{one}, \lambda x : \mathbb{N}.\,\texttt{succ}\, x\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha\times(\alpha\to\alpha)\ \texttt{in}$
$\texttt{pack}\ \langle \nu, \langle (\texttt{snd}\, x)\,(\texttt{fst}\, x), \texttt{snd}\, x\rangle\ \texttt{as}\ \exists\alpha.\alpha\times(\alpha\to\alpha)$

(d) $\lambda P : \exists\alpha.\mathbb{N} \times \alpha \times (\alpha \to \mathbb{N}) \times (\alpha \to \mathbb{B}).$
$\quad\texttt{unpack}\ \gamma, x = P\ \texttt{in}$
$\quad\texttt{let}\ n = \texttt{fst}\,(\texttt{fst}\,(\texttt{fst}\,x))\ \texttt{in}$
$\quad\texttt{let}\ p = \texttt{snd}\,(\texttt{fst}\,(\texttt{fst}\,x))\ \texttt{in}$
$\quad\texttt{let}\ \pi_1 = \texttt{snd}\,(\texttt{fst}\,x)\ \texttt{in}$
$\quad\texttt{let}\ \pi_2 = \texttt{snd}\,x\ \texttt{in}$
$\quad\texttt{pack}\ \langle\gamma, \langle\langle\langle\pi_1\,p, p\rangle, \pi_1\rangle, \pi_2\rangle\rangle\ \texttt{as}\ \exists\alpha.\mathbb{N} \times \alpha \times (\alpha \to \mathbb{N}) \times (\alpha \to \mathbb{B})$

(e) $\lambda P : \exists\alpha.\alpha \times (\alpha \to \mathbb{N}) \times (\alpha \to \mathbb{B}).$
$\quad\texttt{unpack}\ \gamma, x = P\ \texttt{in}$
$\quad\texttt{let}\ p = \texttt{fst}\,(\texttt{fst}\,x)\ \texttt{in}$
$\quad\texttt{let}\ \pi_1 = \texttt{snd}\,(\texttt{fst}\,x)\ \texttt{in}$
$\quad\texttt{let}\ \pi_2 = \texttt{snd}\,x\ \texttt{in}\ \langle\pi_1\,p, \pi_2\,p\rangle$

Question 4: Small-step CBV semantics with existential types

Reduce each of the following terms to a CBV normal form.

(a) $\texttt{pack}\ \langle\mathbb{N}, (\lambda x : \mathbb{N}.\texttt{succ}\,x)\,\texttt{zero}\rangle\ \texttt{as}\ \exists\alpha.\alpha$

(b) $\texttt{unpack}\ \alpha, x = \texttt{pack}\ \langle\mathbb{N} \times \mathbb{N}, \langle\texttt{zero}, \texttt{one}\rangle\rangle\ \texttt{as}\ \exists\alpha.\mathbb{N}\ \texttt{in}\ \texttt{fst}\,x$

(c) $\texttt{unpack}\ \theta, x = \texttt{pack}\ \langle\mathbb{N}, \langle\texttt{zero}, \texttt{one}\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha \times \alpha\ \texttt{in}$
$\quad\texttt{let}\ a_1 = \texttt{fst}\,x\ \texttt{in}$
$\quad\texttt{let}\ b_1 = \texttt{snd}\,x\ \texttt{in}$
$\quad\texttt{let}\ a_2 = \texttt{succ}\,(\texttt{succ}\,b_1)\ \texttt{in}$
$\quad\texttt{let}\ b_2 = \texttt{succ}\,a_1\ \texttt{in}$
$\quad\texttt{pack}\ \langle\mathbb{N}, \langle a_2, b_2\rangle\rangle\ \texttt{as}\ \exists\alpha.\alpha \times \alpha$